

OGUNSOLA TOLUWASE PAUL

Backend Software Engineer | Node.js | TypeScript | Python

Oyo State, Nigeria | Open to Global Remote Roles & Contractor Contracts

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PROFESSIONAL SUMMARY

Backend Software Engineer with 3 years of hands-on software engineering experience building and deploying production applications, distributed systems, and real-time backend architectures using TypeScript, Node.js, and Python. Specialized in PostgreSQL, Redis concurrency control, asynchronous job processing, multi-provider AI infrastructure, and resilient financial workflows. Open to global remote backend engineering opportunities.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend Engineering: Node.js, Express.js, Flask, REST APIs, Asynchronous Processing, Distributed Systems Architecture

Databases & Caching: PostgreSQL, SQLAlchemy, Redis, BullMQ, RQ, SQLite

AI & Real-Time Infrastructure: LLM Orchestration, WebSockets, Socket.IO

Testing & Security: Jest, Pytest, CI/CD, JWT, OAuth, Refresh-Token Rotation, Rate Limiting, RBAC

DevOps & Tools: Docker, Git, GitHub Actions, Supabase, Vercel, Railway, AWS, Gunicorn

KEY ENGINEERING PROJECTS

Kith (Financial Coordination System) | Creator & Lead Backend Engineer

Mar 2026 – Sep 2026 | [GitHub Repository](#) | [Live Demo](#)

- **Financial Integrity & Idempotency:** Designed an append-only transaction ledger to protect historical financial state from mutation; insulated retry-sensitive operations using idempotency keys and duplicate detection, eliminating duplicate ledger entries.
- **Concurrency Safety:** Eliminated check-then-write race windows under concurrent requests by moving race-sensitive workflows into atomic PostgreSQL RPCs, slashing execution time by 35% under concurrent load.
- **Background Processing:** Reduced API response times by 45% by offloading asynchronous workloads to a 9-queue BullMQ architecture with per-queue rate limits and automated recovery scanning, achieving 99.9% delivery reliability.
- **Tenant Isolation:** Enforced multi-tenant security using JWT-to-workspace verification with RBAC, caching authorization states in Redis to reduce database lookups by ~40%.
- **Engineering Standards:** Authored 18 detailed Architecture Decision Records (ADRs) outlining trade-offs in distributed locks, idempotency, and database consistency.

StudyHub (AI & Real-Time Platform) | Creator & Lead Backend Engineer

Jan 2024 – Present | [GitHub Repository](#) | [Live Demo](#)

- **Multi-Provider AI Failover:** Engineered a 6-provider LLM failover system mapping four failure classes (KEY_FAULT, PROVIDER_TRANSIENT, BAD_MODEL, NON_RETRYABLE) to dynamic recovery behaviors, achieving near-100% query uptime.
- **Horizontal Scaling & Distributed State:** Led a horizontal-scaling refactor moving WebSocket presence, scheduler locks, and rate limits from process-local memory to Redis, safely transitioning the service from a single-worker (gunicorn -w 1) bottleneck to multi-worker (gunicorn -w N) scaling.
- **Workload-Matched Task Execution:** Architected a 3-tier task execution system pairing APScheduler with fail-closed Redis locks, RQ for durable retries, and a bounded ThreadPoolExecutor for low-latency AI dispatch, reducing event loop blocking by 80%.
- **Token Rotation & Security:** Implemented DB-backed, SHA-256-hashed refresh token rotation with reuse detection; engineered a 10-second grace window to handle multi-tab races while revoking token families upon stolen-token detection.
- **Automated CI Testing:** Built automated integration test suites in GitHub Actions using fakeredis, Lua scripts via lupa, and isolated in-memory SQLite databases, increasing critical pathway test coverage to 90%+.

FounderSales (AI Sales Coaching Platform) | Creator & Lead Backend Engineer

Feb 2025 – Aug 2026 | [GitHub Repository](#)

- **Multi-Provider AI Fallback Chain:** Architected a 4-provider fallback chain (Cerebras → Groq → Mistral → OpenRouter) with per-key cooldown state shared via Redis across scaled instances, preventing all instances from routing traffic to failing keys instantly.
- **Structured Error Classification:** Replaced string matching with typed ProviderCallError exceptions to distinguish key faults, transient outages, and bad models, stopping provider-wide outages from cooling down healthy keys.
- **Pre-Call Token Gating:** Engineered a pre-call gating layer that reuses prospect research within a 14-day window and logs skip/reuse audit decisions, eliminating token waste on low-stakes calendar events.
- **Cost & Latency Optimization:** Cut LLM API costs by ~60% and reduced latencies from 3.2s to 800ms by consolidating four sequential calls into a single structured payload.
- **Durable Background Queues:** Engineered a 3-queue BullMQ architecture with stable job IDs for deduplication and atomic SQL updates, replacing fire-and-forget IIFEs to eliminate dropped tasks.

EDUCATION

Federal University of Technology, Akure (FUTA) — Oyo State, Nigeria

Bachelor of Science (B.Sc.) in Computer Science — Expected Graduation: 2028